

# Plan for the presentation of my findings

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The main goal is to showcase my findings, and the main findings I want to show is the end result, which is with all the implementations. Meaning that with all of the convolution, heterogeneity, small world topology dynamics.

But first, I need to explain how I started. This should take the shortest amount of time.

## Introduction

Quick introduction about my self, my background (very briefly)

Here I want to talk about that I am doing a spiking reservoir computing framework. I then take this framework and upgrade it multiple implementations that correspond to biological realism, and see how the model performs and what helps.

The end model is a spatially connected and embedded spiking reservoir, with convolutions prior to it.

The goal is not to get highest possible accuracy, but to study how implementing different upgrades helps the model to do better. The accuracy was used just as a simple evaluation.

## First step - defining a simple spiking reservoir computing framework

First slide, show screenshot of their paper title

To do this, we have used BindsNet library, which is built on torch. They had an example script of a reservoir for classifying MNIST dataset. This gave accuracy of around 57%, which is not random guessing, but not too great.

After tweaking some parameters and looking how they decode the information from the reservoir to the readout, I saw a potential upgrade. It was how they decode the temporal information of the reservoir, I changed it and at the end the model performed better, accuracies ranging from 75% to 80%.

Should tell a bit about the change ive done. In that code, they produced a long binary array showing which neurons spiked and which not in every timestep. This made a lot of arrays. I instead of making a seperate array for each timestep, I summed and counted the spikes for each neuron and gave it as a single array to the readout. This improved the accuracy, but what even more improved was to divide the simulation into timebins, where one bin is for example 50ms, and in that bin we have the count of the spikes in 50ms.

So now we have our baseline that we will work on.

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## Making Brunel like reservoir

First slide, start with brunel paper title

- Explain the principle of brunel
  - Sparsly reccurrently connected LIF neurons
  - Two types of neurons, excitatory and inhibitory
  - Noise induced network
- Main hypothesis: Asynchroneous irregular state gives the best performance.
  - Show raster plots of three different states
  - Tell about synchronisity and irregularity
  - Tell about  $g$  and  $\eta$
- Show slide of the results (heatmap plot) youve gotten in previous semester where you tested on iris dataset.
- This means that a specific regime of the network gives the best performance, best information flow. We wish to have a mechanism in the model that would not let the network to run away from this regime.

## Networks self-tuning mechanism

- Tell shortly about CV and rho measurements. And that we can use them to tell the model how to tune  $g$  and  $\eta$  accordingly.
- Show couple gifs as an example of how this self tuning works.
- Tell that when we are using this specific dataset, the model stays in roughly the same regime on all the samples. So this self tuning is not really needed. However, if we would use different data where samples would be vastly different, for example if we would use some sort of time series data or something like that, then some samples would make the network very active, and some samples very quiet, so then perhaps we would need to have this self tuning.
- Now show how you do readout, say that you embed the whole acitivity in vector space, and the idea is that the reseroir will create seperability. Show tSNE plot of this. And give the results

## Introducing spatial dynamics

Well, since the neurons are place spatially.

- Made a simple lattice where I placed all the neurons.
  - Show the EI neuron position. I spread the I neurons randomly
- Toroidal topology, for the connectivity.
  - Every neuron is connected in a gaussian matter.
  - Show plot of outgoing connection of one sample neuron
  - Explain how tunable this is. Epsilon and sigma, these are hyperparameters
- Show some activity plots
- Show tSNE plot
- Show confusion plot

## Introducing convolution

- Introduce feature maps from gabor filters
- Encode these feature maps into spikes
- Show couple of samples of these feature maps
- Talk about how you connect these spiking feature maps into the network
  - Gaussian
  - And how different ways you thought they could be connected
    - Cortical columns
- Show some activity plots with animations
- Show tSNE plot
- Show confusion plot

Of course, different choice of filters and the number of filters has to affect the results. This would be an area to do more research. And of course, multiple layers of convolution should help.

## Introducing heterogeneity and small world topology

- Explain what it means by heterogeneity
  - Show cheng paper that says this
- Explain what it means by small world topology
  - Nice to find a paper that supports this
  - Show outgoing connections plot for this
- Activity plot still do look very much same, so no necessary to see it here.
- Show t-SNE plot
- Show Confusion plot

## Conclusion

Here just show the t-SNE plots together (all of them), and all confusion heatmaps. Just for comparison.

## Next steps ahead

- Testing lognormal connectivity, show a paper that says that this is more biologically realistic, and that my hypothesis is that this will automatically create small world topology.
- And of course, STDP or some other sort of learning
- Energy consumption

